Abderrahim (Rahim) Latreche

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Education

Southern Methodist University | Dallas, TX

M.S. Computer Science, AI/ML Specialization | Expected Dec 2026

B.S. Computer Science (AI/ML Track) and Data Science | Dec 2025 | GPA: 3.9/4.0

Technical Skills

AI / Data: Python, SQL, Snowflake Cortex Analyst/Search/Agents, Snowpark, semantic modeling, RAG, vector search, LangGraph, FAISS, scikit-learn, TensorFlow, Hugging Face

Engineering: JavaScript, React, Next.js, Node.js, Express, FastAPI, REST APIs, Git, Linux, Nginx, systemd, Azure DevOps

Experience

CBRE SmartFM | Data Science Intern

Jun 2026 - Aug 2026 | Dallas, TX | Insights Explorer

- Owned the maintenance-data-to-agent workflow using Snowflake Cortex Analyst, Search, and Agents for conversational analytics.
- Built a semantic model with 10 logical tables, 13 relationships, 60+ facts, dimensions, and metrics, and 12 verified SQL templates for grounded text-to-SQL.
- Fixed VARIANT-derived join keys through staging-view columns to enable building-level rollups; automated version-preserving deployment across 5 LLM variants.
- Built a Snowpark pipeline for irregular Excel tables using structural heuristics and selective LLM assistance; added fallback, quality/cost telemetry, and custom test fixtures. Kept client data inside Snowflake.

CBRE | Software Engineering Intern

Jun 2025 - Aug 2025 | Dallas, TX

- Cleaned and transformed 5M+ HVAC time-series records into features and labeled datasets for fault-detection model development and evaluation against domain-expert rules.
- Built semantic sensor matching with FAISS and embeddings to map natural-language HVAC fault rules to equipment metadata across enterprise portfolios.

Toyota Financial Services | Data Engineer, Sponsored Senior Design

Aug 2025 - Dec 2025 | Plano, TX | Academic project

- Developed Quorra, a React, Python, and REST API query portal with authenticated access, query history, result previews, and downloads.
- Built schema inspection, column summaries, and portfolio analytics workflows for data exploration and SQL query construction.

Southern Methodist University | SURF Research Fellow, NLP

May 2025 - Aug 2025 | Dallas, TX

- Built an AI advising assistant with transformer-based NLP and curated advising datasets; evaluated intent classification and retrieval with precision, recall, and latency.

Selected Projects

Bias Checker | Hugging Face, FastAPI, Next.js

github.com/rahimcantcode/biasCheck

- Built article, sentence, and paragraph predictions with inline passage coloring and inspectable model scores; deployed on a Linux VPS with Nginx, HTTPS, and systemd.

Logistic Regression from Scratch | Python, NumPy, scikit-learn

Optimization, regularization, and dimensionality reduction

- Implemented multiclass logistic regression with gradient ascent, SGD, and Newton's method; explored L1/L2/ElasticNet and PCA retaining 95% variance with 10 of 23 features.